Marker length: what the results show

FYP: synchronization markers for DNAFormer · fine-tuned models, test sets of 1,000 strands per deletion level · 29 Oct 2026

Runs analysed: no markers, marker lengths 4, 5, 6, 7, 8 (periodic) and the length-8 end-block control. **Length 12 is not included:** its results folder in Drive is empty.

38.8% → 52.4%
perfect strands at 8% deletion: no markers → 8-letter markers

4.9% → 2.6%
wrong letters left after decoding at 8% (almost halved)

≈ 50 vs 50
data letters recovered per strand at 8%: no markers vs 8-letter markers (a tie)

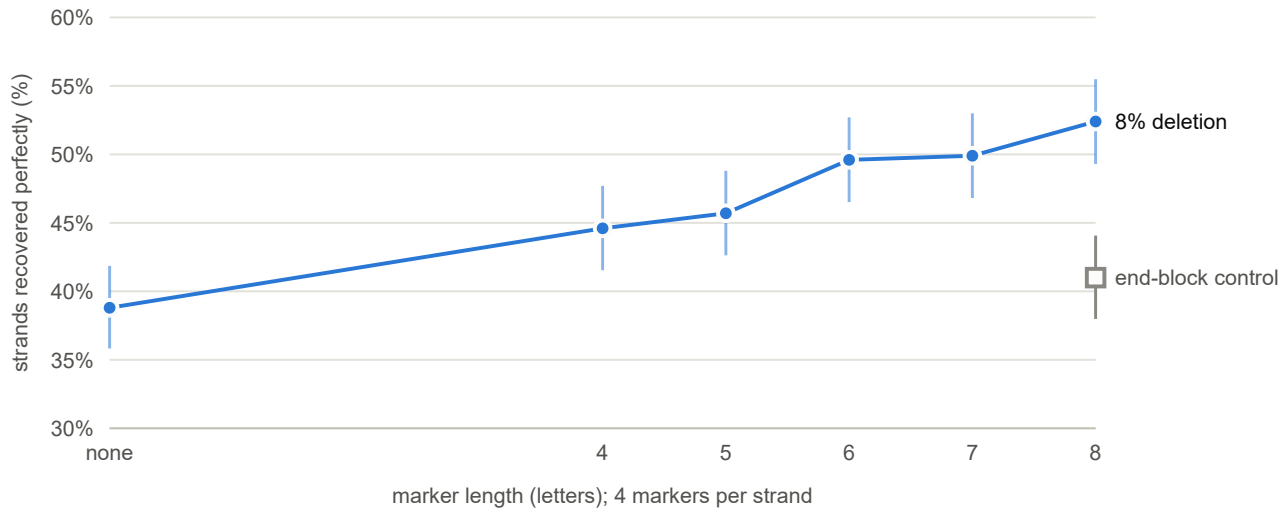
Key insights

1. **Longer markers give more reliable strands.** At 8% deletion, perfect recovery rises steadily with marker length: none 38.8%, 4 → 44.6%, 5 → 45.7%, 6 → 49.6%, 7 → 49.9%, 8 → 52.4%. Wrong letters after decoding fall from 4.9% to 2.6%.
2. **The gain comes from placement, not just from having fewer data letters.** With the same 8-letter markers placed together at the end (end-block control), recovery is only 41.0%, against 52.4% when they are spread out ($p < 0.001$). So the markers carry real synchronization information.
3. **Of the lengths tested, 8 letters gives the most reliable strands.** It has the highest perfect-recovery rate and the lowest edit distance at 8% deletion. Lengths 6, 7 and 8 are close, though: 8 vs 7 (+2.5 points) is not statistically significant.
4. **But markers do not increase the total amount of data recovered.** Each strand holds fewer data letters as markers get longer, and the two effects almost cancel: at 8% deletion every version recovers about 49–52 data letters per strand. At 5% deletion, no markers recovers the most data (120 vs 92).
5. **At 1% and 3% deletion, markers do not matter.** Every version recovers 99.5–100% of strands perfectly. Markers only start to help when deletions are heavy (5% and above), and clearly only at 8%.

> **Bottom line.** Markers make each strand more reliable (fewer errors), and 8 letters is the best length of those tested for reliability. They do not, on their own, increase how much data a strand delivers. Whether they are worth their storage cost depends on the error-correcting code used on top, which is exactly your supervisor's question and the next step.

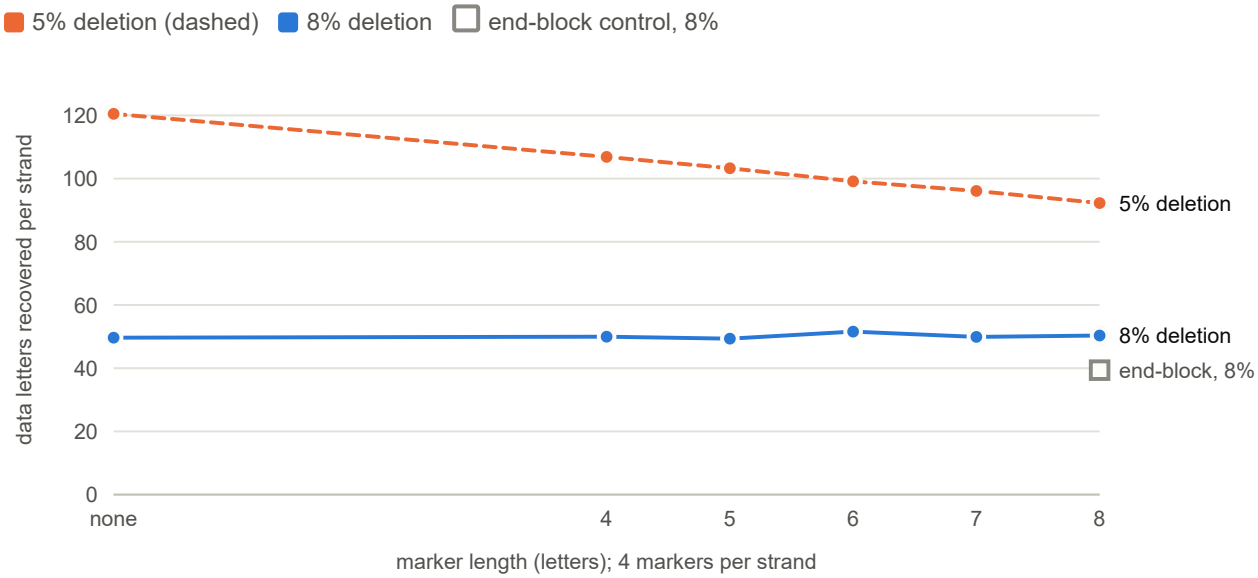
Figures

1. Perfect strand recovery at 8% deletion



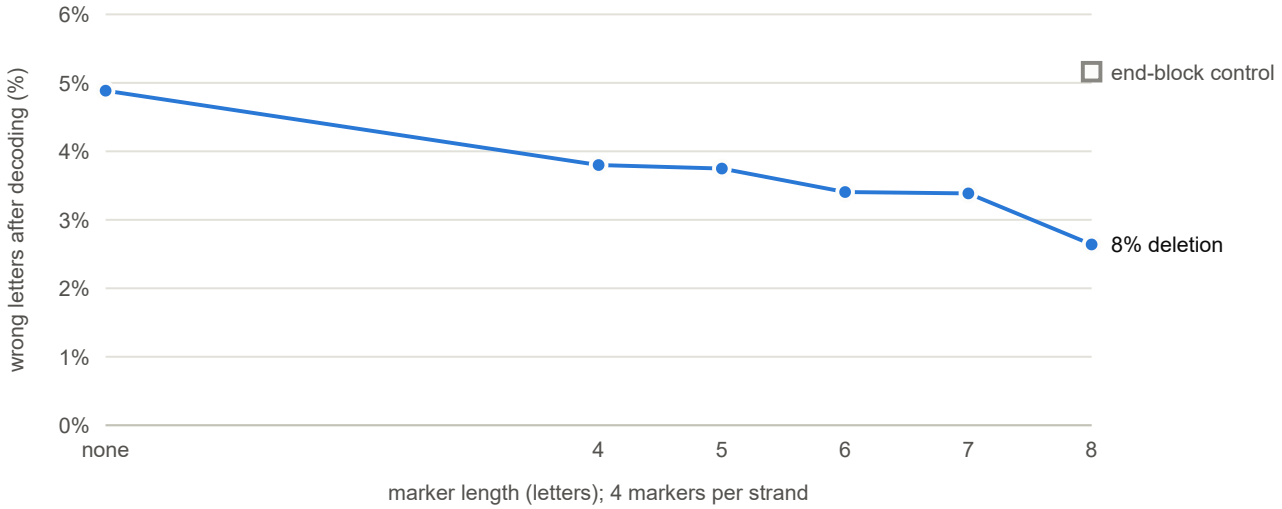
Dots: share of 1,000 test strands recovered with zero errors (data letters only). Bars: 95% confidence intervals. Gray square: 8-letter markers placed together at the end of the strand (control). Recovery grows with marker length; spreading the markers out is what helps.

2. Data letters recovered per strand



Perfect strands × data letters per strand. At 8% the line is flat (≈50 letters for every length): what markers gain in reliability they lose in space. At 5% markers lose data, because almost all strands are recovered anyway.

3. Wrong letters left after decoding (8% deletion)



Share of data letters that are still wrong after DNAFormer (1 − per-letter accuracy). Longer markers almost halve the error rate. Placed at the end, the same markers make it slightly worse than no markers.

4. How much information an ideal error-correcting code could deliver (8% deletion)

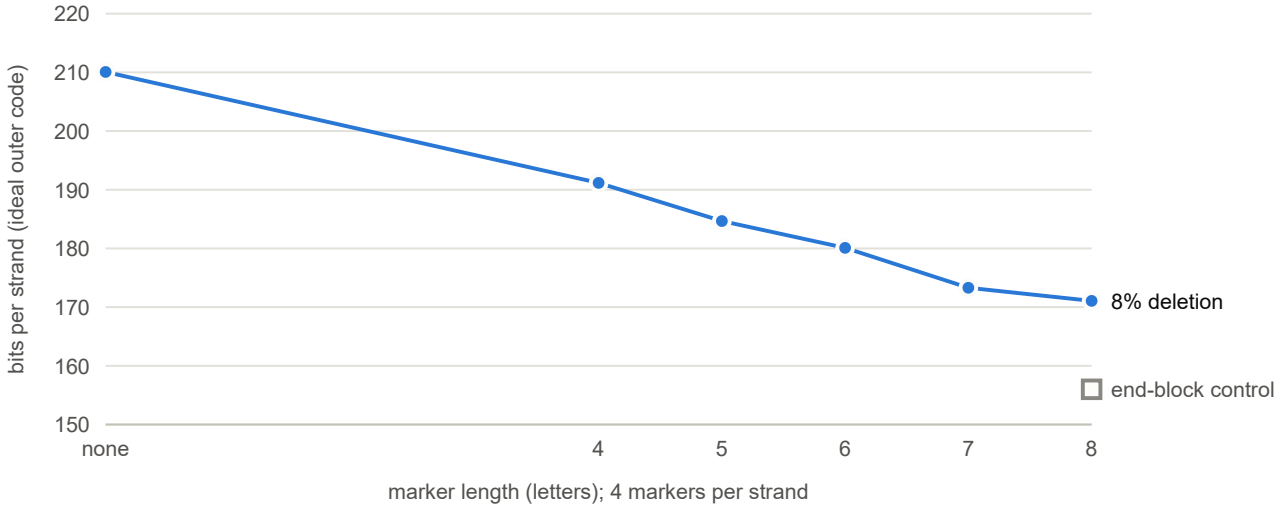


Illustration only: bits per strand = data letters × (2 − H), where H is the information lost to the remaining letter errors, as if an ideal outer code fixed them. It treats leftover errors as independent letter swaps, which is optimistic. Under this estimate no markers still delivers the most information (210 vs 171 bits), so markers only pay off if they make a real code work much better, which has to be tested.

Full results

Version	Marker	Data letters	Marker share	5%: perfect	8%: perfect [95% CI]	8%: NED	8%: wrong letters	8%: data / strand
marker_free	-	128	0.0%	94.1%	38.8% [35.8, 41.9]	0.0202	4.89%	49.7
marker_len04	`CAGT`	112	12.5%	95.4%	44.6% [41.5, 47.7]	0.0183	3.80%	50.0
marker_len05	`TCATG`	108	15.6%	95.6%	45.7% [42.6, 48.8]	0.0182	3.75%	49.4
marker_len06	`CTGTCA`	104	18.8%	95.3%	49.6% [46.5, 52.7]	0.0172	3.41%	51.6
marker_len07	`CTGATCA`	100	21.9%	96.1%	49.9% [46.8, 53.0]	0.0168	3.39%	49.9
marker_periodic	`ACGTCAGT`	96	25.0%	96.1%	52.4% [49.3, 55.5]	0.0148	2.64%	50.3
marker_endblock	`ACGTCAGT (end)`	96	25.0%	95.7%	41.0% [38.0, 44.1]	0.0241	5.16%	39.4

Perfect = strand recovered with zero errors. NED = normalized edit distance (lower is better). Wrong letters = 1 − per-letter accuracy. Data/strand = perfect × data letters. Highlighted: best reliability (8 letters, periodic). At 1% all versions are 100%; at 3%, 99.5–100%. File-level recovery (25-strand files with no error correction) is 0 of 40 for every version at 8%.

Is the difference real? (8% deletion)

Comparison at 8% deletion	Difference	z	p (approx.)	Verdict
Length 8: spaced vs end-block (synchronization)	+11.4 points	5.1	<0.001	clear
Length 4 vs no markers	+5.8 points	2.6	0.008	clear
Length 6 vs length 4	+5.0 points	2.2	0.025	weak
Length 8 vs length 6	+2.8 points	1.3	0.210	not significant
Length 8 vs length 7	+2.5 points	1.1	0.263	not significant
Length 8 vs no markers	+13.6 points	6.2	<0.001	clear

Two-proportion z-test on perfect recovery (1,000 strands each). This ignores the pairing (same strands, same errors), so it is conservative: a paired McNemar test on predictions.csv will be at least as strong. Run that for the final report.

Caveats

- **One training run per version (seed 1).** Differences of 1–3 points (for example 6 vs 7 vs 8) can come from training luck. Retrain the top 2–3 lengths with a second seed before claiming a single best.
- **Length 12 is missing.** It is the point that should show the downside of very long markers. Its results folder is empty, so the run probably did not finish.
- **Simulated data only**, and at deletion rates (5–8%) far above real nanopore reads (~1%).
- **Only one sequence per length.** We showed earlier that sequence matters little for random data, but it was not tested by training.

- **Exact recovery is strict.** One wrong letter counts as a failure, which is why "data recovered" and "wrong letters" are also shown.

Recommended next steps

1. **Finish length 12** to complete the curve.
2. **Run paired tests** (McNemar on predictions.csv) for 8 vs 7, 8 vs 6, and 8 vs no markers.
3. **Second seed** for lengths 6, 7 and 8.
4. **Add an outer error-correcting code** (for example Reed–Solomon across strands) and compare: (a) no markers plus more parity, against (b) markers plus less parity, at the *same total overhead*. This is the fair test of whether markers are worth their cost, and it answers your supervisor's question directly.
5. **Optionally, test 10% deletion**, where markers should help more.

> **What you can claim now:** "Periodic synchronization markers significantly improve DNAFormer's per-strand reliability under heavy deletions (8%), and the benefit grows with marker length up to 8 letters among those tested; the improvement comes from marker placement. However, the gain roughly equals the storage cost of the markers, so their net value depends on the outer error-correcting code."